

EE 451: Communications Systems

Midterm Exam 1 — Study Guide

Exam Date: Thursday, February 26, 2026 (L9) **Duration:** 75 minutes **Coverage:** Chapters 2-3, 7.1-7.2 (Fourier Analysis & Amplitude Modulation)

Exam Format

This exam has two parts, both completed during the 75-minute class period:

Part A — Multiple Choice (20 points, ~15 minutes)

- **20 questions** at 1 point each on **Brightspace**
- Covers conceptual knowledge from the L02, L04, and L07 reading quizzes
- Questions are drawn randomly from the same pool as the **Practice Quiz** (see below)
- Complete Part A first, then close your laptop and begin Part B

Part B — Open-Ended Problems (80 points, ~55 minutes)

- **4 problems** at 20 points each on **paper**
 - **Closed book, closed notes** — a formula sheet is provided
 - Show all work for full credit
 - Problems parallel Homework 1 and Homework 2
 - You may use a scientific calculator
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Practice Quiz (Available Now on Brightspace)

A **Practice Quiz** is available on Brightspace containing all 102 questions from the exam MC pool. This is the same pool that Part A draws from.

- **Unlimited attempts** — take it as many times as you want
 - **Your highest score counts as a homework grade**
 - Questions cover Fourier transforms, AM, SSB/VSB, receivers, and ASK/OOK
 - The best way to prepare for Part A is to take the practice quiz until you can consistently score well
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What to Study

For Part A (Multiple Choice)

The MC questions test conceptual understanding and quick recall. Focus on:

From L02 Reading (Ch 2.1-2.5): - Fourier transform definitions and properties - Time-frequency duality - Dirac delta function properties - Energy vs. power signals - Parseval's theorem

From L04 Reading (Ch 3.1-3.3): - DSB-SC modulation: signal expression, spectrum, bandwidth - Full carrier AM: modulation index, overmodulation - Envelope detection requirements - Power distribution in AM signals

From L07 Reading (Ch 3.6-3.8, 7.1-7.2): - SSB motivation and bandwidth savings - SSB generation methods (filter, phasing) - VSB as a compromise between DSB and SSB - Superheterodyne receiver architecture - Image frequency and its mitigation - ASK/OOK fundamentals

Preparation strategy: Take the practice quiz on Brightspace repeatedly. Review any questions you get wrong.

For Part B (Open-Ended Problems)

The open-ended problems test your ability to set up and solve calculations. Problems are similar in style to your homework assignments. You should be comfortable with the following topics:

Fourier Analysis & Signal Energy - Computing Fourier transforms (especially rect/sinc pairs) - Bandwidth calculations (first null, null-to-null) - Signal energy in time and frequency domains - Parseval's theorem - Time-frequency duality

Amplitude Modulation (Full AM) - Identifying AM signal parameters (A_c , $m(t)$, f_c , μ) - Over-modulation conditions - Power calculations (carrier, sideband, total) - Power efficiency - Spectrum sketching

DSB-SC and SSB - DSB-SC signal expressions and trig identity expansion - Sideband frequencies and RF bandwidth - Power calculations - SSB bandwidth and frequency components - Comparing modulation schemes (bandwidth, power, efficiency)

ASK/OOK and Receivers - OOK signal parameters (bit duration, expression, bandwidth) - Waveform sketching - Superheterodyne receiver calculations (LO frequency, image frequency) - Image rejection and IF trade-offs

Preparation strategy: Redo Homework 1 and Homework 2 problems from scratch. If you can solve the homework without looking at solutions, you are well-prepared for Part B.

Essential Formulas

A formula sheet is provided with the exam. It includes Fourier transform pairs, properties, trig identities, and AM formulas. You do not need to memorize these. However, you should know **how to use** each formula — practice applying them on homework problems.

Key formulas to be comfortable with

Fourier Transform:

$$\begin{aligned} \text{rect}(t/T) & \quad T \cdot \text{sinc}(fT) \\ \text{sinc}(x) &= \sin(x)/(x), \text{ first zero at } x = \pm 1 \end{aligned}$$

Modulation Property:

$$x(t)\cos(2\pi f_c t) \quad \frac{1}{2}[X(f - f_c) + X(f + f_c)]$$

Parseval's Theorem:

$$E = \int |x(t)|^2 dt = \int |X(f)|^2 df$$

AM Signal & Power:

$$s(t) = A [1 + m(t)] \cos(2\pi f_c t)$$

$$P_{\text{total}} = (A^2/2R)(1 + m^2) \quad [\text{general}]$$

$$P_{\text{total}} = (A^2/2R)(1 + m^2/2) \quad [\text{single tone, since } \cos^2 = 1/2]$$

$$= P_{\text{carrier}} / (1 + m^2/2)$$

Bandwidth:

DSB-SC / AM: $2B$

SSB: B

ASK: $2R_b$

Superheterodyne (high-side injection):

$$f_{\text{LO}} = f_{\text{RF}} + f_{\text{IF}}$$

$$f_{\text{image}} = f_{\text{RF}} + 2 \cdot f_{\text{IF}}$$

Time Management (75 minutes)

Phase	Time	What to Do
Part A: Multiple Choice	~15 min	Answer 20 MC questions on Brightspace
Problem 1: Fourier	~12 min	FT, bandwidth, energy, Parseval's
Problem 2: AM Power	~14 min	Parameters, overmodulation, power, efficiency, spectrum
Problem 3: DSB-SC/SSB	~14 min	Trig expansion, sidebands, power, comparison table
Problem 4:	~15 min	OOK waveform, bandwidth, receiver
ASK/Superheth		frequencies, Q trade-off
Review	~5 min	Check answers, verify units
Total	75 min	

Common Mistakes to Avoid

Fourier Transform: - Forgetting the factor of T in $X(f) = AT \text{sinc}(fT)$ - Confusing first null ($1/T$) with null-to-null bandwidth ($2/T$) - Mixing up energy signals vs. power signals

AM Power: - Forgetting the factor of $1/2$ in power: $P = A^2/(2R)$ for a sinusoid - Not checking for overmodulation before calculating power - Confusing μ (modulation index) with $m(t)$ (message signal) - Forgetting to include resistance R in power calculations

DSB-SC / SSB: - Forgetting the $1/2$ factor from the trig identity $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ - SSB bandwidth is B , not $2B$

ASK / Superheterodyne: - Confusing baseband bandwidth (R_b) with null-to-null RF bandwidth ($2R_b$) - Sign errors in image frequency calculation - Confusing the RF preselector filter (rejects image) with the IF filter (selects channel)

Self-Assessment Checklist

Before the exam, make sure you can:

Fourier Analysis: - ☐ Compute FT of rectangular pulse and identify sinc parameters - ☐ Find first null and null-to-null bandwidth - ☐ Calculate signal energy in time domain - ☐ Verify energy using Parseval's theorem - ☐ Explain time-frequency duality

Amplitude Modulation: - ☐ Extract A_c , A_m , f_c , f_m , μ from an AM equation - ☐ Check for overmodulation ($\mu > 1$?) - ☐ Calculate carrier, sideband, and total power - ☐ Calculate efficiency η - ☐ Sketch magnitude spectrum with labeled components

DSB-SC and SSB: - ☐ Write DSB-SC expression and expand with trig identity - ☐ Identify USB and LSB frequency components - ☐ Calculate DSB-SC power (sum of component powers) - ☐ Determine SSB-USB components and bandwidth - ☐ Compare DSB-SC, SSB, and AM (bandwidth, power, efficiency)

ASK and Superheterodyne: - ☐ Calculate bit duration and null-to-null bandwidth for OOK - ☐ Sketch OOK waveform from a bit sequence - ☐ Calculate LO frequency and image frequency - ☐ Explain the image problem and preselector mitigation - ☐ Explain why higher IF helps image rejection but hurts channel selectivity

Study Resources

Homework (Most Important!)

- **Homework 1:** Problems 3, 6, 7 (Euler's formula, signal energy, power signals)
- **Homework 2:** Problems 1-3, 5-7, 9-10 (DSB-SC, AM power, bandwidth, OOK, superheterodyne, AM system design)
- **Homework 2 Bonus 1:** SSB bandwidth and power (extra credit on HW, but good exam practice)
- Redo these problems from scratch — don't just re-read solutions

Practice Quiz

- Available on Brightspace — take it multiple times
- Covers all MC topics; highest score counts as homework

Lecture Notebooks

- **L01:** Complex numbers and phasors
- **L02:** Signal analysis, energy/power, I/Q representation
- **L03:** Fourier analysis, rect/sinc, bandwidth
- **L04:** Parseval's theorem, spectral density
- **L05:** AM modulation, DSB-SC, envelope detection
- **L06:** ASK/OOK, spectral analysis, superheterodyne
- **L07:** SSB, Hilbert transform, VSB

Textbook

- Chapter 2.1-2.5: Fourier Transform and properties
 - Chapter 3.1-3.3: AM, DSB-SC
 - Chapter 3.6-3.8: SSB, VSB
 - Chapter 7.1-7.2: ASK, OOK
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Good luck!